

Experiment no.

Aim: To study DC motor starters.

Theory:

In case of DC motors, the back emf developed in the armature winding is directly proportional to the speed of rotation. When the motor is just connected to DC supply, the back emf is zero since speed is zero causing a current through winding.

R_a , the resistance of armature winding is quite low of the order 0.5 to 1 ohm. This current can be limited to a safe value by adding an external resistance in the form of starters. A DC motor starter consists of a properly grounded starting resistance. It is also provided with protective devices.

Three-Point Starter:

A three-point starter for a DC shunt motor having starting resistance graded in nine steps and provided with no-volt and overload release to protect the motor from different possible hazards. The three terminal of motor L, Z and A are connected respectively to the positive terminal, shunt field terminal and armature terminal. The starter arm is held at the OFF position during the off state.

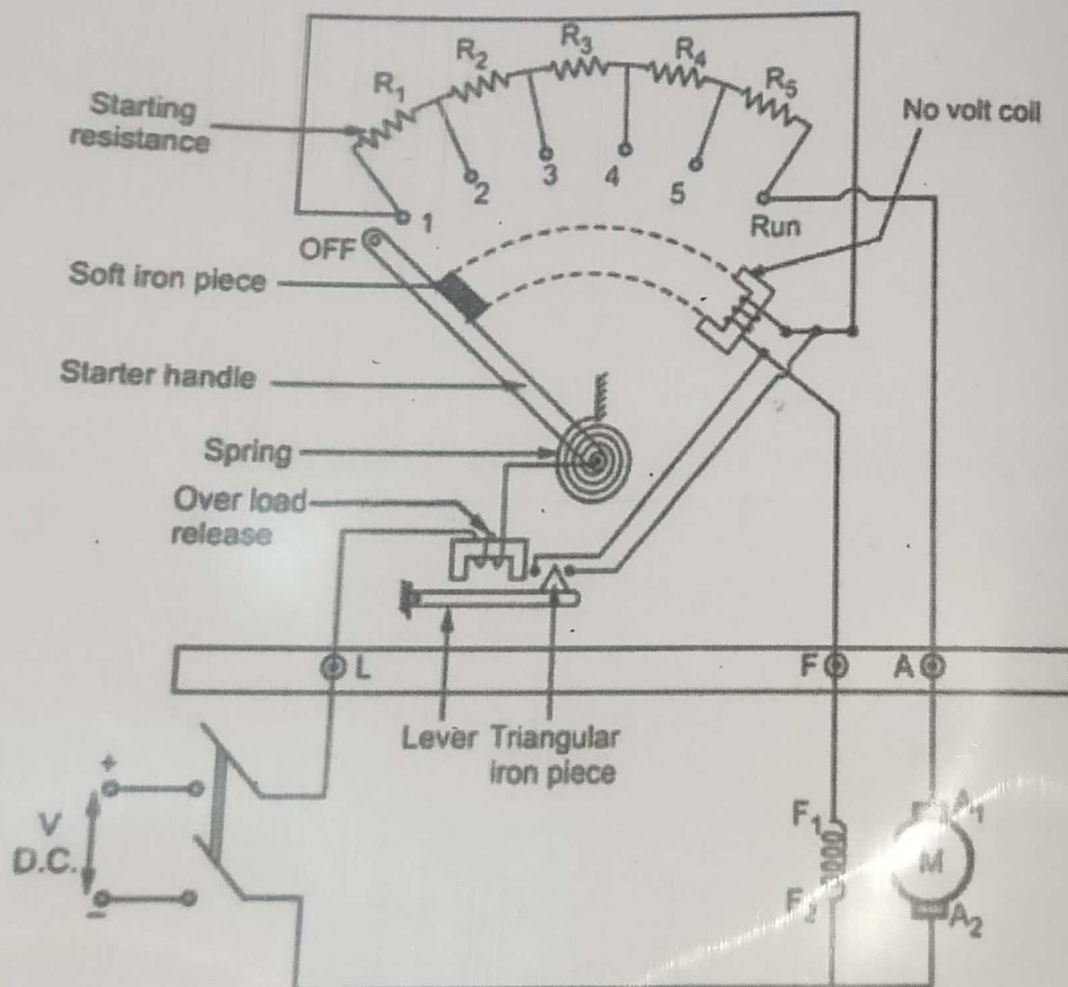
The motor can be started by the field main switch provided on supply side and then shifting the starter arm, on the first step which includes the complete resistance of all the nine steps in the armature circuit, thereby redirecting the stability.

$$I_s = \frac{V}{R + \sum R_s}$$

where $\sum R_s$ is the sum of resistance of all steps in the starter only.

No-Load release

The no volt release consists of an electromagnet with proper winding on it. Hence in the event of a power failure or an accidentally opening the field circuit.

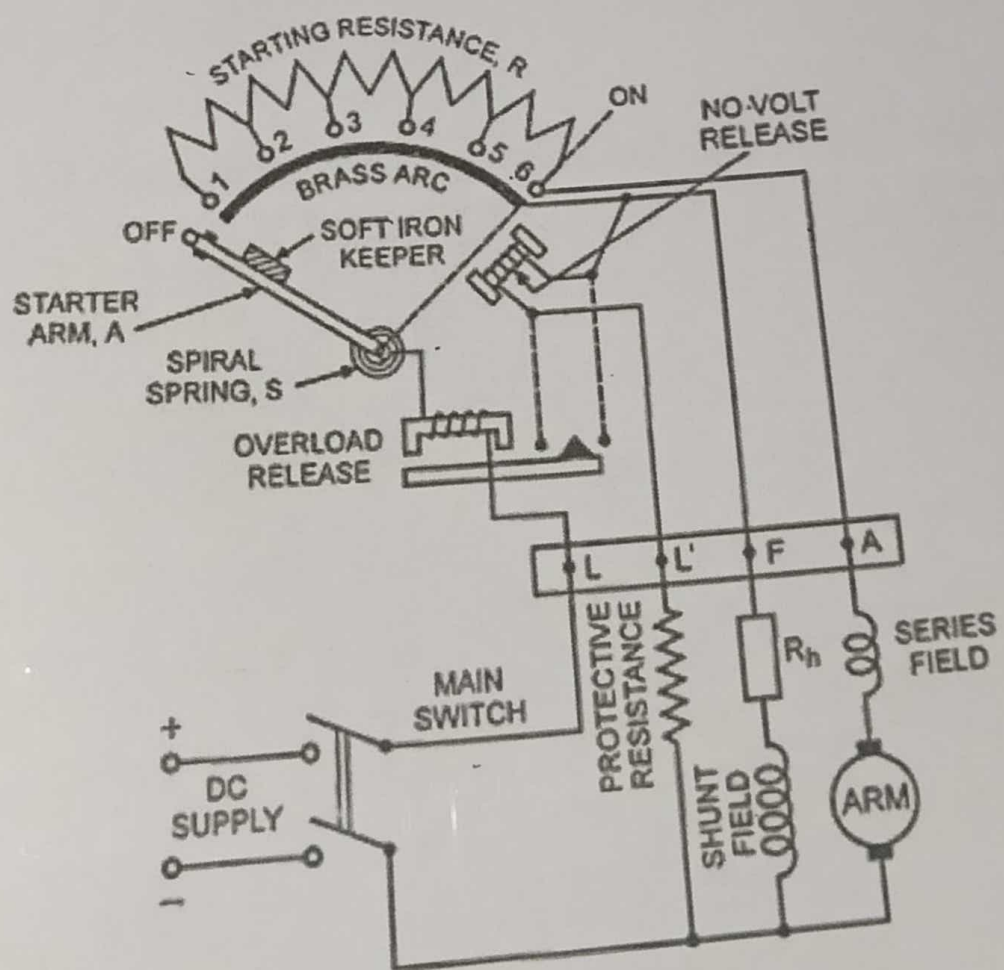


3 Point Starter Circuit Diagram

Overload release

It consists of an electromagnet with a proper winding on it. It is connected in series with the armature circuit so that current flowing in the electromagnetic winding is equal to the armature current. If the motor becomes overloaded drawing an overload current from line then electromagnet will release.

Four-Point starter: The basic difference in the circuit of four-point starter as compared to the three-point starter is that the holding coil has been removed from the shunt field circuit and has been connected directly to the mains with a current limiting resistance in series such as an arrangement forms three parallel circuits. One consists of overload release starting resistance and armature winding resistance, second consisting of variable resistance of shunt field and third having holding coil.



4 Point Starter Circuit Diagram